Curriculum Vitae
Shui-Sen Zhang, Ph. D.

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Education

University of Science and Technology of China (USTC) 09/2021-07/2026 (expected)
& High Magnetic Field Laboratory, HFIPS, Chinese Academy of Sciences (CAS)

- Ph. D. in Condensed Matter Physics
  Thesis: *The asymmetric spin transport and the associated dynamics of antiferromagnets*
  Advisor: Prof. Ding-Fu Shao & Prof. Haifeng Du

Fudan University (FDU) 09/2016-07/2021

- B. S. in Physics
  Thesis: *The measurement based on ferromagnetic resonance and the determination of the spin precession angle in Py/Pt*
  Advisor: Prof. Yizheng Wu

Research Interests

- Transport properties and dynamics in antiferromagnets.
- Analysis and simulations of the motion of the topological magnetic structure.

Publications

[1] **S.-S. Zhang**, Z.-A. Wang, B. Li, W.-J. Lu, M. Tian, Y.-P. Sun, H. Du, and D.-F. Shao, *Deterministic Switching of the Néel Vector by Asymmetric Spin Torque*, **Phys. Rev. Lett.** 136, 96702 (2026).

[2] **S.-S. Zhang,** Z.-A. Wang, B. Li, Y.-Y. Jiang, S.-H. Zhang, R.-C. Xiao, L.-X. Liu, X. Luo, W.-Ji. Lu, M. Tian, Y.-P. Sun, E. Y. Tsymbal, H. Du, and D.-F. Shao, *X-type stacking in cross-chain antiferromagnets*, **Newton** 1(4), 100068 (2025). (**cover article**)
Preview: J. Železný, *A new type of antiferromagnetic stacking*, **Newton** 1, 100109 (2025).

[3] L. Li, D. Song, W. Wang, L. Kong, **S. Zhang**, N. Wang, S. Zhang, M. Tian, J. Zang, Y. Liu, and H. Du, *Electrically writing a magnetic heliknoton in a chiralmagnet*, **Nature Materials**. (2026).

[4] Y. Zhang, W. Liu, M. Shi, **S. Zhang**, S. Qiu, Y. Wu, J. Jiang, H. Zhang, H. Han, K. Wang, D. Shao, Z. Zi, C. Ma, H. Du, M. Tian, S. Wang, and J. Tang, *Enhanced Curie temperature and room-temperature 50-nm skyrmions achieved in hexagonal ferromagnet $Mn_5Ge_{3.2}$ synthesized via a high-pressure method*, **Science China Physics, Mechanics & Astronomy** 69, 247511 (2026).

[5] Z.-A. Wang, B. Li, **S.-S. Zhang**, W.-J. Lu, M. Tian, Y.-P. Sun, E. Y. Tsymbal, K. Wang, H. Du, and D.-F. Shao, *Giant uncompensated magnon spin currents in X-type magnets*, **Chinese Physics Letters** 42(11): 110713 (2025). (**Express Letter**)

[6] D. Song, W. Wang, **S. Zhang**, Y. Liu, N. Wang, F. Zheng, M. Tian, R. E. Dunin-Borkowski, J. Zang, and H. Du, *Steady motion of 80-nm-size skyrmions in a 100-nm-wide track*, **Nature Communications** 15, 5614 (2024).

Professional Skills

- **DFT calculation**: Quantum ESPRESSO, VASP, PAOFLOW, Wannier90.
- **Numerical methods/Simulations:** Python, Mathematica, MATLAB, MuMax3.
- **Document Preparation:** LaTeX, MS Office.

Selected Posters

2026.2 *Deterministic Switching of the Néel Vector by Asymmetric Spin Torque*, Future Asia Conference on Spintronics and Magnetic Recording Technologies (FAST2026), Phuket, Thailand.

2025.6 *X-type antiferromagnetic and asymmetric spin torque*, Workshop on Altermagnets and School on Magnetic Symmetry, Shanghai, P. R. China **(Best Poster Award)**.

2025.6 *X-type stacking in cross-chain antiferromagnets*, 2025 IEEE Magnetics Society Joint Nanjing-Hong Kong Workshop on Spintronics, Nanjing, P. R. China **(Best Poster Award)**.

2024.11 *X-type antiferromagnetic stacking for spintronics*, CPS Youth Spintronics Workshop VII, Hefei, P. R. China **(Best Poster Award)**.

Synergistic Activities

- **Journal Referee:** Physical Review B, Physical Review Applied.

Reference

1. Prof. Ding-Fu Shao
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2. Prof. Haifeng Du
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3. Prof. Bo Li
School of Physics, Xi'an Jiaotong University, China
E-mail: libphysics@xjtu.edu.cn

4. Prof. Yizhou Liu
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